

NOT ALL NUKES ARE CREATED EQUAL
THEATER NUCLEAR FORCES AND PERCEPTIONS OF ESCALATION

Advisors: Madeline Berzak and Decker Eveleth

XLab Risk and Security Working Group

Summer 2026

Abstract

Modern warfighting places particular emphasis on theater nuclear forces, as great powers tailor their nuclear strategies for regional, rather than global, contingencies. Advocates push for expanding US theater nuclear forces, while critics caution that crossing the nuclear threshold could lead to uncontrolled escalation. Furthermore, traditional models of escalation are often ineffective for framing theater conflict. This research aims to address whether certain attributes of theater nuclear weapons systems shape escalation perception independently of effect. I also propose a novel model of inadvertent and accidental escalation: the escalation carousel. Through vignette-based survey methods, I intend to test whether escalation perceptions follow a categorical logic driven by the nuclear-conventional distinction, or a graduated logic driven by launch site and delivery vehicle, with implications for ongoing theater force structure debates.

1. Introduction

Nuclear weapons redefined what it means to engage in a limited war. Prior to the nuclear age, a conflict was considered limited if leaders pursued restrictive, manageable objectives and deployed only a fraction of military resources, stopping short of total mobilization. Nuclear weapons inverted this definition, as states were no longer constrained by limited resources but instead possessed the ultimate weapon. Restraint became the defining feature of limited war, as fear of mutually assured destruction (MAD) drove great powers to avoid direct wars with each other. Today, competition and friction between major powers manifests in a different domain, through theater conflicts and proxy wars.

Despite the taboo on the use of nuclear weapons that characterized the MAD paradigm, military strategists have actively explored how states might use nuclear weapons in combat situations, particularly as conflict increasingly concentrates at the theater level. Contemporary nuclear strategy debates are hyperfocused on theater nuclear force structure: modernization, miniaturization, and advancements in weapons technology have produced nuclear weapons that are smaller, yield-variable, and highly precise, raising questions about battlefield applications. Modeling these scenarios, however, has become increasingly difficult when scholars look to traditional and linear frameworks in international relations theory built for a different era and characterized by the Cold War's bipolar security environment. Traditionally, seminal nuclear strategist Herman Kahn is most often linked to escalation, as he developed the escalation ladder in his 1965 book *On Escalation*.¹ The ladder is a powerful visual metaphor for modeling conflict that enables strategic planners and policymakers to forecast linear crisis developments. Although intended as a theoretical framework, Kahn's ladder is a visualization tool that came to function as

¹ Herman Kahn, *On Escalation: Metaphors and Scenarios* (New York: Frederick A. Praeger, 1965).

a de facto planning framework within the American national security establishment; the image of the ladder, in many ways, became akin to doctrine. However, the ladder lacks nuance and does not adequately capture the evolutions in modern warfighting, much of which occurs in the “gray zone” between peacetime and active conflict. Kahn’s metaphor rests on untested assumptions about how policymakers perceive escalation, leaving planners with an unreliable tool from which to craft informed strategy and policy. Although the latter third of Kahn’s ladder grapples with nuclear options, contemporary nuclear debates exist within an evolved tripolar security environment between the US, Russia, and China, with non-kinetic attacks and dual-capable delivery systems complicating the escalation calculus. Within Kahn’s ladder, theater nuclear use was positioned as a stepping stone to an all-out nuclear war between great powers, yet today, theater nuclear use is considered almost more urgent of a risk than full-scale war.

The integration of emerging technology into military doctrine and force structure further complicates Kahn’s metaphor, and recent scholarship on potential for nuclear use with low-yield or, “theater,” nuclear weapons challenges the longstanding norm of nuclear non-use. Since 1945, nuclear weapons have existed in a siloed category; a categorical redline grounded in the existence of a nuclear taboo and in deterrence theory. However, as the United States pours funding and resources into modernizing its nuclear arsenal and thinking about how tripolar competition changes this calculus, conversations about theater nuclear use have increased in frequency and severity, threatening this perceived redline. Drawing on Erik Lin-Greenberg’s 2023 *Evaluating Escalation* study and means-based framework, I examine whether means-based logic crosses the nuclear threshold.² Through a scenario-based survey experiment, I explore whether weapons attributes, specifically launch location and delivery vehicle, shape perceptions

² Erik Lin-Greenberg, “Evaluating Escalation: Conceptualizing Escalation in an Era of Emerging Military Technologies,” *The Journal of Politics* 85, no. 3 (2023), <https://doi.org/10.1086/723974>.

of escalation independently of yield, and how American policymakers perceive the escalatory weight of their own arsenal.

2. Relevance

Theater nuclear force structure remains an area of active policy debate and development. The US, alongside Russia and China, is currently developing and modernizing various theater nuclear weapons systems, including the low-yield, nuclear sea-launched cruise missile (SLCM-N) and the B61-12 gravity bomb. Proponents of expanding US theater nuclear forces argue that the current US arsenal is insufficient to provide credible deterrence and respond proportionally to the range of modern conflict. Leading policy voices on this side of the debate contend that expanding the arsenal would enable *flexible* deterrence, whereas maintaining only strategic nuclear weapons leaves the US with a “suicide or surrender” dilemma.³ The US is behind competitors in this domain, but policy advocates argue that investing in theater nuclear forces would demonstrate a credible, usable deterrent to allies and adversaries alike.⁴

Contemporary debates about theater nuclear weapons reflect a shift from viewing nuclear weapons primarily as a signaling tool for deterrence to a potential battlefield asset. Conversations around escalation also reflect this shift. Policymakers and scholars frequently ask which US systems will have what effect on an adversary, but the escalation literature neglects the reverse question: how do US policymakers and strategists perceive their own weapons and capabilities. If US policymakers are set on developing theater nuclear forces strategies, research on understanding escalation perceptions is required to create the most informed strategy with the

³ Vipin Narang, “The Discrimination Problem: Why Putting Low-Yield Nuclear Weapons on Submarines Is So Dangerous,” *War on the Rocks*, February 8, 2018, <https://warontherocks.com/discrimination-problem-putting-low-yield-nuclear-weapons-submarines-dangerous/>.

⁴ Amato, “Not Imaginary.”

highest likelihood of success. This study informs real-time debates by examining which weapons American policymakers actually believe usable and why. Existing escalation models also fail to capture the mechanisms most relevant to theater weapons. Deliberate-escalation ladders fail to model inadvertent and accidental pathways the US government fears. This study also proposes an updated visual metaphor, the escalation carousel, to more effectively model the escalation mechanisms most relevant to theater nuclear forces.

3. Literature Review

3.1 The Usability of Nuclear Weapons

A distinct lack of nuclear weapons use characterizes the last 80 years. The tradition of non-use persists despite nuclear modernization trends of weapons becoming more precise, yield-variable, and technologically advanced, or in other words, more “usable.” Nuclear states frequently tout the necessity of nonproliferation frameworks and while they seemingly believe in the value of this tradition, increasingly serious debates about theater nuclear use and posture changes weaken the strength of Nina Tannenwald’s “nuclear taboo.” As policymakers consider the idea of limited nuclear use, a growing body of literature further examines what factors decision-makers care about when debating whether to push the button.

This flirtation with nuclear usability is not new. The Eisenhower era in the 1950s was characterized by strong support for the US nuclear arsenal, especially tactical nuclear weapons.⁵ Economic concerns drove Eisenhower’s stance, and he maintained that active conventional forces were much more costly than relying on nuclear weapons.⁶ The investment and development of tactical nuclear forces during the early Cold War was an effort to

⁵ Andrew Glass, “Eisenhower Defends Use of Nuclear Weapons, March 16, 1955,” *Politico*, March 16, 2019, <https://www.politico.com/story/2019/03/16/eisenhower-defends-use-of-nuclear-weapons-march-16-1955-1224003>.

⁶ Stephen I. Schwartz, “The Costs of U.S. Nuclear Weapons,” *Nuclear Threat Initiative*, September 30, 2008, <https://www.nti.org/analysis/articles/costs-us-nuclear-weapons/>.

“conventionalize” nuclear weapons, exemplified by Eisenhower’s famous statement: “Now, in any combat where these things can be used on strictly military targets and for strictly military purposes, I see no reason why they shouldn’t be used just exactly as you would use a bullet or anything else.”⁷ The US nuclear arsenal expanded tremendously under the Eisenhower administration, but despite the increase of available weapons and a strategy widely communicated as massive retaliation, the US never used nuclear weapons.⁸

Tannenwald’s 1999 thesis *The Nuclear Taboo: The United States and the Normative Basis of Nuclear Non-Use* attempted to explain the disconnect between the Eisenhower era buildup and the norm of non-use. Tannenwald argued that a normative prohibition against using nuclear weapons is behind nuclear non-use. Her work established the categorical claim in the literature that nuclear weapons occupy a category of their own—weapons of mass destruction, distinct from conventional arms, and decision makers treat them accordingly.⁹ Tannenwald’s taboo argument shaped scholarship on nuclear non-use, but the next generation of research suggests the taboo is not as strong as Tannenwald originally posited.

T.V. Paul argued in his 2010 piece *Taboo or Tradition? The Non-Use of Nuclear Weapons in World Politics* that the term “taboo” is too strong and overrepresentative of the phenomenon, proposing instead the tradition of non-use.¹⁰ Scott Sagan, Ben Valentino, and Daryl Press further challenge the taboo thesis with their 2013 public opinion survey. Their data found that public

⁷ Glass, “Eisenhower Defends Use of Nuclear Weapons.”

⁸ Jeffrey Lewis, “Point and Nuke,” *James Martin Center for Nonproliferation Studies*, September 12, 2018, <https://nonproliferation.org/point-and-nuke/>; Richard M. Leighton, *Strategy, Money, and the New Look, 1953–1956*, History of the Office of the Secretary of Defense, vol. 3, ed. Alfred Goldberg (Washington, DC: Historical Office, Office of the Secretary of Defense, 2001), https://history.defense.gov/Portals/70/Documents/secretaryofdefense/OSDSeries_Vol3.pdf.

⁹ Nina Tannenwald, “The Nuclear Taboo: The United States and the Normative Basis of Nuclear Non-Use,” *International Organization* 53, no. 3 (1999): 433–68. <http://www.jstor.org/stable/2601286>.

¹⁰ T. V. Paul, “Taboo or Tradition? The Non-use of Nuclear Weapons in World Politics,” *Review of International Studies* 36, no. 4 (2010): 853–63, <http://www.jstor.org/stable/40961956>.

support for nuclear use hinges less on any absolute moral prohibition than on relative military utility; when nuclear weapons outperform conventional ones, Americans significantly support their use. Those who opposed nuclear use, however, did so not out of moral abhorrence, but out of fear of setting a negative precedent.¹¹ Sagan et al.'s findings challenged the categorical framing of nuclear weapons. That public support for nuclear use tracks military stakes rather than moral prohibition becomes especially consequential in contemporary debates about theater nuclear forces, where high precision, limited-use scenarios are precisely for what they are designed.

Sagan et al.'s survey work captured a public willing to support nuclear use when militarily advantageous in the early 2010s. Yet David M. Allison's, Stephen Herzog's, and Lauren Sukin's 2026 piece *Too Brutal for War: Comparing Rationales for Weapon Taboos*, shows the opposite result.¹² Through a conjoint survey experiment, Allison et al. explore public attitudes towards use of various weapons, finding that even when nuclear weapons are more effective, respondents prefer a non-nuclear, less effective strike 76% of the time.¹³ Allison et al.'s work lends new experimental support to the taboo thesis and categorical argument. Taken together, these juxtaposing results reflect an active debate over the strength of the nuclear threshold.

¹¹ Daryl G. Press, Scott D. Sagan, and Benjamin A. Valentino, "Atomic Aversion: Experimental Evidence on Taboos, Traditions, and the Non-Use of Nuclear Weapons," *The American Political Science Review* 107, no. 1 (2013): 188–89, <http://www.jstor.org/stable/23357763>.

¹² David M. Allison, Stephen Herzog, and Lauren Sukin, "Too Brutal for War: Comparing Rationales for Weapon Taboos," *Journal of Peace Research* 63, no. 4 (2026): 690–704, <https://doi.org/10.1093/jopres/xjag001>.

¹³ Allison et. al, "Too Brutal for War."

To study potential public support for nuclear weapons use, scholars have moved beyond surveys and tested the nuclear usability question via wargames.¹⁴ Wargaming became the method to gather data unavailable from observed history.¹⁵ In his paper *Would U.S. Leaders Push the Button?* Reid Pauly details a series of wargames with national security and defense officials, finding that participants very infrequently used nuclear weapons due to fear of reputational deterioration.¹⁶ Pauly's findings lend experimental support to the tradition of nuclear non-use, with reluctance rooted in the reputational costs Tannenwald and Paul theorize.

Andrew Reddie and Bethany Goldblum's wargaming work builds off of Pauly's study, testing the usability question when low-yield nuclear weapons are on the table. In a series of wargames published in 2022, Reddie and Goldblum studied player responses when given no nuclear options, strategic nuclear weapons, and/or low yield nuclear weapons. Through the study, *Evidence of the Unthinkable: Experimental Wargaming at the Nuclear Threshold*, Reddie and Goldblum find that the presence of low-yield options increases the odds of nuclear use by as much as 24%.¹⁷ Reddie and Goldblum's findings raise usability concerns and further support the conclusion that participants feel some connection to a tradition of nuclear non-use, but not at the expense of failing a military objective. Regarding escalation and perceptions, Reddie and Goldblum's work provides experimental support for the proposition that players perceive these weapons differently based on yield, writing, "Our analysis also finds support for the proposition

¹⁴ Philip Jan Schäfer and Joseph Verbovsky, "Wargames Are Experiments: Wargames as Experimental Methods in International Relations," *Zeitschrift für Außen- und Sicherheitspolitik* (2026), <https://doi.org/10.1007/s12399-026-01107-3>.

¹⁵ Erik Lin-Greenberg, Reid B. C. Pauly, and Jacquelyn G. Schneider, "Wargaming for International Relations Research," *European Journal of International Relations* 28, no. 1 (March 2022): 83–109, <https://doi.org/10.1177/13540661211064090>.

¹⁶ Reid B. C. Pauly, "Would U.S. Leaders Push the Button?: Wargames and the Sources of Nuclear Restraint," *International Security* 43, no. 2 (Fall 2018): 151–192, https://doi.org/10.1162/isec_a_00333.

¹⁷ Andrew W. Reddie and Bethany L. Goldblum, "Evidence of the Unthinkable: Experimental Wargaming at the Nuclear Threshold," *Journal of Peace Research* 60, no. 5 (2022).

that low-yield nuclear weapons substitute for their high-yield counterparts – suggesting that even in nuclear conflict, players internalize distinctions in the use of different types of force.”¹⁸ The key difference between Pauly’s work and that of Reddie and Goldblum’s game is that the latter isolated yield as the independent variable. This study builds on the concept of isolating a characteristic as an independent variable.

Across nearly three decades of scholarship, researchers have utilized different methods and studied different populations to explain what makes nuclear weapons usable (or not). The literature has moved from Tannenwald’s foundational argument of the nuclear taboo to a fragile tradition of non-use to a contested norm. Throughout this evolution, the nuclear threshold consistently functions as a categorical and uniform boundary. Now, nuclear policy debates are returning to usability questions from the 1950s; back then the nuclear boundary was treated as flexible – a graduated logic – and only now do those questions confront a boundary that four decades of theory have rendered firmly categorical.

3.2 The Contemporary Debate: Theater Nuclear Forces

The relevance and strategic value of theater nuclear forces challenge that boundary; contemporary debates on US nuclear strategy reflect a wavering commitment, as force structure arguments increasingly presume that some nuclear options are less escalatory than others. Theater nuclear forces promise flexibility. Since the 1990s, Russia has maintained a distinct numerical advantage in theater nuclear weapons, a category that formal arms control agreements never restricted.¹⁹ This asymmetry of capabilities, characterized as a “deterrence gap,” is a key

¹⁸ Reddie and Goldblum, “Evidence of the Unthinkable.”

¹⁹ Anatoly Zagorski, “Executive Summary,” [United Nations Institute for Disarmament Research \(UNIDIR\)](https://unidir.org/files/conferences/pdfs/executive-summary-a-zagorski-eng-0-325.pdf), accessed July 5, 2026, <https://unidir.org/files/conferences/pdfs/executive-summary-a-zagorski-eng-0-325.pdf>; Steven Pifer, “Arms Control Options for Non-Strategic Nuclear Weapons,” *Brookings*, July 1, 2026, <https://www.brookings.edu/articles/arms-control-options-for-non-strategic-nuclear-weapons/>.

motivation for US expansion of theater nuclear forces.²⁰ The 2026 National Defense Strategy prioritizes theater nuclear weapons as a means to deter and potentially defeat two nuclear peers and to manage escalation.²¹ Those in favor of expanding theater nuclear forces argue that the US needs calibrated response options tailored to specific theaters and conflicts, but critics caution that any nuclear use, however limited, risks uncontrolled escalation.

Former Deputy Director for Strategic Stability Greg Weaver maps the current theater nuclear force landscape, identifying seven key attributes effective theater nuclear forces must possess to provide credible deterrence and support certain missions, and noting where the US military is lacking.²² Former Director of Nuclear Deterrence Policy Paul Amato further urges the US to expand its theater nuclear forces, arguing that these low-yield weapons are the backup plan in case deterrence fails.²³ With its current arsenal, US deterrence capability depends on strategic nuclear forces and conventional weapons. While the US maintains a decisive conventional advantage against Russia, Russia's theater nuclear forces remain superior. In the event Russia leverages its theater nuclear weapons for coercive value, the US lacks a timely, proportional response.²⁴

Russia is not the US's only consideration for expanding theater nuclear forces. Deterring in Europe and Asia is a strategic priority, and one of which the US is incapable with its current

²⁰ Paul Amato, "Not Imaginary: The Deterrence Gap is Real and America Needs Low-Yield Nukes," War on the Rocks, June 26, 2026,

<https://warontherocks.com/not-imaginary-the-deterrence-gap-is-real-and-america-needs-low-yield-nukes/>.

²¹ US Department of War, *2026 National Defense Strategy* (Washington, DC: Department of War, January 23, 2026), 15, <https://media.defense.gov/2026/Jan/23/2003864773/-1/-1/0/2026-national-defense-strategy.pdf>; Heather Williams and Nicholas Adamopoulos, "Trump's New Nuclear Architecture for Modernization and Arms Control," *Center for Strategic and International Studies* (CSIS), March 13, 2026,

<https://www.csis.org/analysis/trumps-new-nuclear-architecture-modernization-and-arms-control>.

²² Greg Weaver, "The Imperative of Augmenting US Theater Nuclear Forces" (Washington, DC: Atlantic Council, April 2025),

<https://www.atlanticcouncil.org/wp-content/uploads/2025/03/The-imperative-of-augmenting-US-theater-nuclear-forces.pdf>.

²³ Amato, "Not Imaginary."

²⁴ Ibid.

force structure. Matthew Kroenig argues for augmenting theater nuclear weapons on the basis of deterring two nuclear peers and assuring allies. The US currently has no deployed theater nuclear forces in Asia and an insignificant amount deployed in Europe.²⁵ Furthermore, US allies are doubting the US's commitment to extended deterrence.²⁶ Kroenig argues that deploying theater nuclear forces in Europe and Asia is a physical signal of commitment to allies and deterrence to adversaries.²⁷

The proponents' case itself rests on assumptions about escalation. Weaver's focus on characteristics implies a graduated logic of escalation; weapons with specific characteristics occupy a different escalatory position from other weapons, and certain theater nuclear weapons are better suited to missions intended to limit escalation.²⁸ Amato's argument goes further, arguing that possessing limited nuclear options for contingency purposes implies a belief in controlling nuclear escalation, and should things escalate, a belief in being able to fight and win nuclear wars.²⁹

Weapons attributes themselves have become a central point of contention. Vipin Narang, an academic and former government official, has held in his academic work that nuclear postures

²⁵ Matthew Kroenig, "U.S. Strategic Force Posture for the Third Nuclear Age," Ronald Reagan Presidential Foundation & Institute, July 6, 2023, <https://www.reaganfoundation.org/reagan-institute/publications/us-strategic-force-posture-for-the-third-nuclear-age>.

²⁶ U.S. Congress, Senate, Committee on Armed Services, To Receive Testimony on Strategic Competition in an Unconstrained, Post-New START Treaty Environment, 119th Cong., 2nd sess., February 3, 2026, <https://www.armed-services.senate.gov/hearings/to-receive-testimony-on-strategic-competition-in-an-unconstrained-post-new-start-treaty-environment>.

²⁷ Kroenig, "U.S. Strategic Force Posture."

²⁸ Greg Weaver, "The Imperative of Augmenting US Theater Nuclear Forces" (Washington, DC: Atlantic Council, April 2025), <https://www.atlanticcouncil.org/wp-content/uploads/2025/03/The-imperative-of-augmenting-US-theater-nuclear-forces.pdf>. Weaver identifies the following attributes: survivability, continuous deployment in both theaters - Europe and Asia, variation in warhead yield and fusing options, operationally relevant delivery timelines, high likelihood of penetrating adversary defenses with limited strikes, effectiveness against full range of likely targets, and ability to augment US hedge capability and meet strategic targeting requirements.

²⁹ Henrik Stålhane Hiim, 2024. "The Last Atomic Waltz: China's Nuclear Expansion and the Persisting Relevance of the Theory of the Nuclear Revolution." *Contemporary Security Policy* 45 (2): 239–64. doi:10.1080/13523260.2023.2291258.

deter unequally. In 2018, the US government was debating whether to equip the Trident submarine-launched ballistic missile (SLBM) with a low-yield warhead. Narang argued in *The Discrimination Problem: Why Putting Low-Yield Nuclear Weapons on Submarines Is So Dangerous* that equipping traditionally strategic nuclear weapons with low-yield warheads creates a scenario in which adversaries cannot distinguish whether a limited or city-destroying strike is on its way. Narang notes that early warning systems cannot determine warhead yield, and writes, “Proponents of these capabilities have elided one key problem: how the adversary may perceive and react to their use.”³⁰ Austin Long responds to Narang, arguing in *Discrimination Details Matter: Clarifying an Argument About Low-Yield Nuclear Warheads* that the so-called discrimination problem already exists, predating the low-yield proposal, and that context will temper perceptions.³¹ The US and UK share missiles, so if a Trident SLBM were launched at an adversary, the adversary could not distinguish whether it were a US launch, a UK launch, or a joint strike. Furthermore, Long reasons that if an adversary has already conducted a limited strike, a Trident launch in response would likely be interpreted as proportional.³² Narang’s position is not categorical opposition to flexible deterrence. During his tenure at the Pentagon from 2022 to 2024, he articulated a more competitive posture toward US adversaries, but his 2018 debate with Long centered on how various attributes may invite misperception.

Discussions about response options should traditional deterrence fail reflect both a commitment to some form of the tradition of nuclear non-use—the US wants response options that are not as destructive as strategic nuclear weapons, implying strategic nuclear weapons as unusable—and an acceptance that crossing the nuclear threshold may become necessary to avoid

³⁰ Narang, “The Discrimination Problem.”

³¹ Austin Long, “Discrimination Details Matter: Clarifying an Argument About Low-Yield Nuclear Warheads,” *War on the Rocks*, February 16, 2018, <https://warontherocks.com/discrimination-details-matter-clarifying-argument-low-yield-nuclear-warheads/>.

³² Long, “Discrimination Details Matter.”

all out nuclear war. But proponents of arms control and nonproliferation warn of the dangers posed by this paradox. The categorical position, that is, arguing that any nuclear use crosses a line, appears less in recent scholarship than in the arms control policy community, though its intellectual roots lie in the nuclear revolution tradition. Jervis's 1989 book, *The Meaning of the Nuclear Revolution: Statecraft and the Prospect of Armageddon*, is a foundational argument warning about uncontrolled escalation. Jervis argued that the greatest risk for nuclear war comes not from deliberate action but from humans' inability to control a spiraling crisis.³³ Tannenwald reexamines her earlier arguments in her 2018 *Foreign Affairs* piece, *The Vanishing Nuclear Taboo?*, noting how President Trump's nuclear rhetoric and having flexible yield options threatens a strong nuclear threshold.³⁴

Officials have opposed theater nuclear force expansion in government as well. In 2018, then Secretary of Defense James Mattis testified before the House Armed Services Committee stating that he does not believe "there is any such thing as a tactical nuclear weapon. Any nuclear weapon used any time is a strategic game changer."³⁵ In 2020, then Chairman of the House Armed Services Committee Adam Smith opposed funding of the W76-2 warhead – a low-yield warhead mounted on the Trident SLBM. Smith argued, "The Administration's decision to deploy the W76-2 warhead remains a misguided and dangerous one. The deployment of this warhead does nothing to make Americans safer. Instead, this destabilizing deployment further increases the potential for miscalculation during a crisis. Validating the utility of so-called 'low-yield' nuclear weapons in 'winning' a nuclear war adds to the growing pressures of a nuclear arms

³³ Robert Jervis, *The Meaning of the Nuclear Revolution: Statecraft and the Prospect of Armageddon* (Ithaca, NY: Cornell University Press, 1989), 238-247.

³⁴ Nina Tannenwald, "The Vanishing Nuclear Taboo? How Disarmament Fell Apart," *Foreign Affairs* 97, no. 6 (November/December 2018), [foreignaffairs.com](https://www.foreignaffairs.com).

³⁵ Congressional Research Service, "Nonstrategic Nuclear Weapons," CRS Report No. RL32572, version 36 (Washington, DC: Congressional Research Service, January 17, 2019), https://www.congress.gov/crs_external_products/RL/PDF/RL32572/RL32572.36.pdf.

race.”³⁶ Today, those against force expansion further argue that modernizing the US nuclear arsenal is already a massive undertaking, and expanding theater forces further strains budgetary resources.³⁷ Others object on arms control grounds, warning that every investment in expansion diverts political capital from repairing a regime already in collapse.³⁸

Modern debates reflect a lack of a common perception of escalation. While some value attributes, implying a graduated logic of escalation, others see any nuclear use as the categorical line to be crossed or upheld. Those that do raise concerns about escalation perception do so in relation to adversary perceptions, not domestic. The literature has largely left unasked what US policymakers make of these new nuclear options; estimating adversary perceptions is valuable, but understanding domestic perceptions of one’s own arsenal is a prerequisite for knowing what signal these weapons are meant to send in the first place. While experimental work studying the nuclear threshold has varied whether limited nuclear options are available and measured whether players use them, it has not systematically varied the attributes of those options nor treated perceived escalation itself as the outcome of interest.

3.3 Escalation Theory and Dynamics

Understanding the escalatory implications of various weapons systems directly informs which weapons American policymakers categorize as usable. Although the dynamic of escalation goes back centuries—Carl von Clausewitz described belligerents driving each other

³⁶ House Armed Services Committee Democrats, “Smith Criticizes the Administration’s Deployment of W76-2 Low-Yield Nuclear Warheads,” press release, February 4, 2020, <https://democrats-armedservices.house.gov/2020/2/smith-criticizes-the-administration-s-deployment-of-w76-2-low-yield-nuclear-warheads>.

³⁷ Adam Mount, “The Case Against New Nuclear Weapons,” Center for American Progress, May 4, 2017, <https://www.americanprogress.org/article/case-new-nuclear-weapons/>.

³⁸ Tyler Bowen, “Restrain and Hedge: A New U.S. Nuclear Strategy for a Two-Peer World,” *War on the Rocks*, May 15, 2026, <https://warontherocks.com/restrain-and-hedge-a-new-u-s-nuclear-strategy-for-a-two-peer-world/>.

toward extremes—escalation theory emerged during the Cold War era as strategists grappled with what exactly makes something escalatory. Today, 20th century physicist Herman Kahn is the scholar universally linked to escalation theory due to his highly memorable visualization.

Kahn's *On Escalation* introduced the “escalation ladder,” a perennially popular metaphor, in 1965.³⁹ Kahn's 44 rungs describe conflict from sub-crisis maneuvering to great power wars. Kahn rejected the idea that nuclear weapons precluded strategic thought altogether and instead theorized that escalation could still be deliberately managed in stages. Kahn's ladder models key concepts in escalation theory, including horizontal and vertical escalation, escalation dominance, and thresholds or “firebreaks.”⁴⁰ The ladder offers a tool that still has utility today for thinking about pathways, and its grouping of rungs under categorized thresholds is valuable for thinking about strategic options. However, it is too strict and simplistic to be an effective model for evaluating risks associated with the modern threat landscape.

Kahn's ladder primarily emphasizes one mechanism of escalation: deliberate. However, one cannot fall up a ladder; the linear model cannot fully explain inadvertent or accidental escalation—situations highly likely to occur in modern conflict. Inadvertent escalation refers to intentional actions that are unintentionally escalatory, often because the action crosses an adversary threshold unbeknownst to the acting party. Accidental escalation occurs when military forces take intentional actions not intended by leadership, either due to misunderstanding intention or disregarded orders. Inadvertent escalation poses the most acute and feared escalation risk in today's security landscape, but both escalation mechanisms cannot be deterred, only managed through signaling, perception, and communication.⁴¹ The ladder itself is an incomplete

³⁹ Kahn, *On Escalation*.

⁴⁰ Ibid.

⁴¹ Eric A. Wallace, Mathias D. Boggs, Samuel F. Gailliot, and Adam D. Williams, “Exploring Multilayer Network Models to Build a Scientific Basis for Integrated Deterrence: Final Report,” SAND2025-12012R (Albuquerque,

model, unable to account for all pathways to escalation, and critics go further still, questioning whether escalation can be modeled as a controllable process at all.

In the 1960s and 1970s, Thomas Schelling and Robert Jervis rejected the linear logic of Kahn's ladder and its illusion of perfect control. Schelling argued that escalation is a "competition in risk-taking" — a "slippery slope" rather than a ladder, where power lies in the manipulation of uncertainty rather than calculated steps.⁴² Jervis rejected Kahn's rational actor assumption, arguing that leaders' perceptions, not objective reality, drive international relations; interpretation of escalation varies from leader to leader, and signals are rarely interpreted correctly.⁴³

These decades-old critiques still inform the escalation debate, especially when considering the role of theater nuclear forces. In the last ten years, scholars and policymakers have colloquially titled Russia's nuclear strategy "escalate to deescalate," or E2D. The basic premise is as follows: an actor launches a limited nuclear strike to shock and frighten the adversary into conceding to war termination on the escalating actor's terms; E2D is, essentially, an extreme form of compellence.⁴⁴ When E2D is discussed in the Russian context, the strategy is labeled as risky and frightening. But some scholars, like former Belfer Center Senior Fellow Kevin Ryan, hold that while perhaps under a different name, the US maintains E2D as a strategy

NM: Sandia National Laboratories, 2025); Forrest E. Morgan, Karl P. Mueller, Evan S. Medeiros, Kevin L. Pollpeter, and Roger Cliff, "Dangerous Thresholds: Managing Escalation in the 21st Century" (Santa Monica, CA: RAND Corporation, 2008), <https://www.rand.org/pubs/monographs/MG614.html>.

⁴² Thomas C. Schelling, *The Strategy of Conflict* (Cambridge, MA: Harvard University Press, 1960), 187-205.

⁴³ Robert Jervis, "Introduction." In *Perception and Misperception in International Politics*: New Edition, REV-Revised. Princeton University Press, 1976. <https://doi.org/10.2307/j.ctvc77bx3.5>.

⁴⁴ Daniel R. Post, "Escalating to De-escalate with Nuclear Weapons: Research Shows It's a Particularly Bad Idea," *Bulletin of the Atomic Scientists*, February 9, 2024, <https://thebulletin.org/2024/02/escalating-to-de-escalate-with-nuclear-weapons-research-shows-its-a-particularly-bad-idea/>.

as well.⁴⁵ Others argue that E2D is and never was Russia's strategy at all.⁴⁶ Popular as E2D may be among the superpowers, scholars Daniel Post and Marina Henke argue that a limited nuclear strike may not be a winning strategy.⁴⁷ The strategy overpromises success and invites more risk than reward, and Henke's survey work finds that escalation is far likelier to lead to retaliation rather than anticipated conciliation.⁴⁸

This third wave of scholarship on escalation theory also examines cross-domain escalation, emerging technologies, and regional dynamics. Conventional wisdom suggests that emerging technologies make escalation significantly worse and faster. Caitlin Talmadge rejects this conclusion, arguing that emerging technology is at best an intervening variable but not sufficient or necessary to alone lead to escalation, an argument emphasizing strategy over technological determinism.⁴⁹ Talmadge's study, however, examines technical factors in conventional escalation dynamics, not nuclear, relegating technical factors to intervening variable status.

Emerging technologies have further blurred the traditional conventional-nuclear boundary, raising new questions about escalation modeling. Erik-Lin Greenberg examines this intersection of emerging technology and escalation and challenges fundamental frameworks. Lin-Greenberg's work returns to a question raised by original nuclear strategist Bernard Brodie in the 1960s. Brodie recognized in 1966 that attribute,

⁴⁵ Post, "Escalating to De-escalate with Nuclear Weapons."

⁴⁶ Michael Kofman and Anya Loukianova Fink, "Escalation Management and Nuclear Employment in Russian Military Strategy," *War on the Rocks*, September 19, 2022, <https://warontherocks.com/escalation-management-and-nuclear-employment-in-russian-military-strategy-2/>.

⁴⁷ Post, "Escalating to De-escalate with Nuclear Weapons;" Marina E. Henke, Dino Kolonic, and Felix Lemmer, "Russian Roulette Revisited: Does a Limited Nuclear Strike Constitute a Winning Strategy?" *Contemporary Security Policy* 47, no. 2 (2026): 655–79, <https://doi.org/10.1080/13523260.2026.2624653>.

⁴⁸ Henke, "Russian Roulette Revisited."

⁴⁹ Caitlin Talmadge, 2019. "Emerging Technology and Intra-War Escalation Risks: Evidence from the Cold War, Implications for Today." *Journal of Strategic Studies* 42 (6): 864–87. <https://doi.org/10.1080/01402390.2019.1631811>.

means-based questions about theater nuclear weapons had been “hardly at all discussed publicly,” likely reflecting a lack of strategic planning at the classified level.⁵⁰ Traditional literature on escalation theory, including Kahn’s ladder, ranks various actions as more or less escalatory based on the action’s effect. Lin-Greenberg proposes an alternative, a means-based framework, ranking actions based on how the attack was delivered, finding that officials perceive actions more physically present and visible as more escalatory.⁵¹ On his proposed, revised ladder, officials perceive ground attacks as most escalatory and cyberattacks as least escalatory. Lin-Greenberg’s study, however, does not interrogate how nuclear weapons fit into these emerging technology dynamics.

Scholars are also examining what these modern escalation dynamics might look like at the theater level, research that perhaps most closely informs modern theater nuclear forces debates. In *Would China Go Nuclear? Assessing the Risk of Chinese Nuclear Escalation in a Conventional War with the United States* Talmadge puts forth a perception-driven theory of escalation. In theoretically playing out a conventional war between the US and China, she argues that even if nuclear use was originally unintended, factors like dual-use bases and conventional-nuclear integration change how the adversary interprets escalation.⁵² Talmadge’s study highlights material characteristics of forces as influencing perceptions of escalation. More broadly, escalation scholarship is moving from analyzing an action to analyzing the physical attributes of involved forces.

⁵⁰ Bernard Brodie, “Introduction.” In *Escalation and the Nuclear Option*. Princeton: Princeton University Press, 1966, 9.

⁵¹ Lin-Greenberg, “Evaluating Escalation,” 85, no. 3 (2023).

⁵² Caitlin Talmadge, “Would China Go Nuclear? Assessing the Risk of Chinese Nuclear Escalation in a Conventional War with the United States,” *International Security* 41, no. 4 (Spring 2017): 50–92, https://doi.org/10.1162/ISEC_a_00274.

Despite recent scholarship challenging the efficacy of limited nuclear strikes, US support for theater nuclear forces continues to grow, implying a belief that with tailored weapons, escalation is controllable, yet which attributes drive perceptions of escalation remains largely unexamined. Henke's survey varies target, leaving other attributes still empirically untested. Because governments have decided theater systems are necessary for deterrence, the assumption that tailored weapons can control escalation is no longer theoretical but embedded in policy. If that assumption is to be tested rather than trusted, research must independently examine the attributes believed to make escalation controllable.

4. Theoretical Framework: The Escalation Carousel

4.1 Mapping Escalation Dynamics onto the Carousel

Current models cannot capture these relationships between attributes, perception, and policy. In light of these gaps, I present a new model of escalation that offers increased flexibility and analytical power. Most models focus on deliberate escalation, but deliberate escalation is not the most relevant mechanism for scenarios involving theater nuclear weapons. The US government is increasingly concerned about inadvertent or accidental escalation with theater nuclear weapons;⁵³ I propose a metaphor to model these two mechanisms: the escalation carousel. A carousel has several features that lend themselves to modeling these mechanisms of escalation: rotation, whose momentum is external to any rider; horses that rise and fall as part of normal motion; height that is meaningful only relative to other riders; a cycle that recurs without accumulating; and a variety of exit options.

⁵³ James M. Acton, "Inadvertent Escalation and the Entanglement of Nuclear Command-and-Control Capabilities," Carnegie Endowment for International Peace, October 29, 2018, <https://carnegieendowment.org/posts/2018/10/inadvertent-escalation-and-the-entanglement-of-nuclear-command-and-control-capabilities>; James Johnson, "Inadvertent Escalation in the Age of Intelligence Machines: A New Model for Nuclear Risk in the Digital Age," *European Journal of International Security* 7, no. 3 (August 2022): 337–359, <https://doi.org/10.1017/eis.2021.23>.

The classic escalation ladder is a rigid metaphor built to reflect the time in which Kahn created it. Designed to model state-to-state conflict, the ladder implies that escalation is driven by and between states—Kenneth Waltz’s second level of analysis.⁵⁴ It exists in a vertical geometric plane in which actors keep climbing toward the precipice of disaster and implies a one-way direction of escalation. I propose that modern escalation dynamics can be better explained through a systemic lens—Waltz’s third level of analysis.

Waltz argues that war occurs because we exist in anarchy; no global government exists.⁵⁵ Anarchy governs the carousel: the international system stays in constant rotation even when riders—sovereign states—take no action to spin it. The carousel captures the cyclical, systemic nature of escalation. While inherently circular, a carousel is not a spiral hurtling toward disaster; instead, it mirrors the system’s cycle of escalation and temporary peace, in which factors outside of direct state intent drive both escalation and deescalation. Furthermore, just as modern conflicts involve multiple states and non-state actors, a carousel has multiple riders. Theater conflicts and proxy wars involve entire state blocs. The carousel’s ability to model interconnectedness and alliances reflects real world situations of horizontal escalation and inadvertent consequences of escalation; actions taken by one rider to impact another rider will likely all impact the carousel. The carousel describes escalation as a function of tempo and momentum rather than level or rung. The faster the carousel spins, the riskier the system and the higher the chances of inadvertent or accidental escalation.

The carousel metaphor also introduces a new element: escalation capacity. A carousel is a pay-to-play experience, implying that escalation inherently requires a form of entry into the system, some threshold one must cross to gain the capacity to escalate. In today’s international

⁵⁴ Kenneth N. Waltz, *Man, the State, and War: A Theoretical Analysis* (New York: Columbia University Press, 2001), <https://doi.org/10.7312/walt12537>.

⁵⁵ Kenneth N. Waltz, *Man, the State, and War*.

system, that threshold has been lowered—the democratization and proliferation of weapons has lowered the price of admission, and the barrier is not exclusive to the nuclear realm. The number of actors on the carousel has grown accordingly, and the capacity of smaller-scale non-state actors to provoke and engage in escalation with great powers has increased.

The carousel metaphor acknowledges that states do play a significant role in escalation dynamics. In the metaphor, escalation occurs because the carousel's riders have the capacity to influence the intensity of its speed. An actor's position in the system (the height of its horse) is meaningful only relative to its adversary's position at that specific moment in time. In this way, the carousel captures escalation dynamics as discrete moments in time relative to specific actors. An actor's escalatory potential fluctuates over time, and so does its manner of signaling escalation. In the US, signaling changes by administration; depending on who is in power, escalatory potential is signaled differently, including the question of whether to signal escalatory capacity at all. Factors within states, below the systemic level, dictate whether a horse is moving up or down.

The carousel also models off ramps that other metaphors fail to include. For Kahn's ladder model, de-escalation is a separate, deliberate act, where actors climb down rung by rung; on a carousel, de-escalation requires no separate act, because rising and falling are inherent to the ride itself. A carousel slows down and sometimes even stops. Riders can get off anytime—even jumping off is an option, although riders may injure themselves in the process. The carousel thus retains the realistic human attribute of choice: actors can choose when and how to get off the carousel, and the metaphor represents the consequences of doing so.

I've established the carousel metaphor to model conventional escalation dynamics. With a hint of absurdity, one can extend the model to the nuclear world. In this metaphor, escalation

works because you can threaten to stop the ride. Imagine each rider on our carousel is a suicide bomber with bombs strapped to their chest. They threaten use to try and change the intensity of the ride, but no rider wants to outright blow up the carousel. The balance between bluffing and weapons use is what keeps the nuclear carousel turning: threats change the intensity of the ride only because every rider knows the bombs are real, and the ride continues only because every rider knows that detonation would destroy the carousel itself. But the greatest danger on a spinning platform is not that a rider chooses to detonate. It is that a bomb goes off independent of anyone's deliberate choice, due to a jolt of the ride, a stumble between horses, or a rider misreading an adversary's reach for the trigger. The intangible systemic factors are what raise the danger of inadvertent and accidental escalation.

4.2 Speed, Risk, and Decision Time: The Carousel for Policymakers

The escalation ladder endures because policymakers find it easy to use. Rungs are intuitive, and decision makers can map events onto them quickly. However, that ease comes at the cost of leaving out variables that more accurately represent the international system. The carousel gives up some of that simplicity to gain a dimension—speed alongside direction. I've established the relationship between the speed of the carousel and the riskiness of the system; as speed increases, risk increases. The carousel does not predict what any actor will do. Its predictive value comes from identifying when the system is entering conditions where inadvertent actions and accidents become likely. The carousel tracks a single policy-relevant variable: decision-making time.

Speed represents the tempo of interaction: the rate at which events demand a response, or the inverse of decision-making time. This variable is also relevant for policymakers. Indicators like the frequency of diplomatic exchanges or shifts in the tone or tempo of official

communications are observable proxies for the carousel's speed, or ways of measuring the rate of interaction without having to guess at intent. But speed belongs to the system rather than to any rider. Although a singular actor may influence the speed, slowing the carousel down requires a collaborative, multi-actor approach. When the indicators show the ride accelerating, decision windows contracting, the conditions for accidents forming, the carousel also gives policymakers tools to slow it.

The model prescribes risk reduction measures, such as arms control, confidence-building measures, and crisis communication, as policy tools to manage the carousel's speed and avoid inadvertent and accidental escalation. By giving policymakers indicators to watch and tools to act, the carousel earns the same usability that made the ladder endure, without leaving out the system spinning beneath the riders.

4.3 Theater Nuclear Forces on the Carousel

Returning to the realm of the absurd, the carousel can also model theater nuclear forces. If strategic nuclear weapons are bombs strapped to the riders' chest, then theater nuclear weapons are many tiny bombs—some so small they may not even be seen. States do not intend for use of a small bomb to destroy the carousel, only to shake things up to adjust the course, or speed, of the ride on the user's terms. This scenario follows the “escalate to deescalate” logic, or an extreme circumstance of compellence. But a tiny bomb is only worth carrying if the rider holding it believes they can successfully detonate it without drastically increasing the speed or destroying the ride altogether. In the real world, stationing theater nuclear forces in any theater sends a signal of commitment to allies and adversaries, as well as an implicit signal of self-confidence; by deploying theater nuclear forces, a state broadcasts its key assumption that

with those weapons, they can control escalation dynamics. On the carousel, this is a rider announcing they can detonate a small bomb *and* keep the ride turning.

5. Methods

To assess how emerging nuclear and defense policymakers perceive the escalatory potential of theater nuclear weapons, I designed an original vignette-based survey experiment that draws on wargaming and scenario methods. Participants read a one-page scenario describing a border crisis in Kaliningrad and are asked to advise the President of the United States on a response. Plausibility is essential to scenario-based research, as participants disengage from scenarios they find unbelievable.⁵⁶ This vignette is reminiscent of contemporary international security challenges. As Russia's war in Ukraine continues into its fourth year, NATO–Russia crisis dynamics remain an active policy debate, and ongoing tensions over Kaliningrad and the Suwałki Gap provide the scenario with a sense of urgency. US debates over bolstering theater nuclear forces in Europe, driven by fears of Russian theater nuclear use further reinforces the vignettes' credibility.

The survey isolates two independent variables: launch site and delivery vehicle. Yield and target type are well-established drivers of escalation perceptions.⁵⁷ Launch site and delivery vehicle, by contrast, remain understudied, and offer more nuanced insight into how attributes of the weapons system itself, rather than its destructive effects, shape perceptions of escalation. These variables are also more relevant to force structure debates. To measure their impact, I present participants with nine response options that vary along these two dimensions. Across all nine response options, the vignette holds warhead yield, target, and casualties constant; both

⁵⁶ SJSU School of Information, "Vignettes in Research: A Practical Method for Studying Real-World Decisions in LIS," SJSU, January 19, 2026,

<https://ischool.sjsu.edu/post/vignettes-research-practical-method-studying-real-world-decisions-lis>.

⁵⁷ Daryl G. Press, Scott D. Sagan, and Benjamin A. Valentino, "Atomic Aversion: Experimental Evidence on Taboos, Traditions, and the Non-Use of Nuclear Weapons," *The American Political Science Review* 107, no. 1 (2013): 188–89, <http://www.jstor.org/stable/23357763>.

nuclear options carry the same low-yield warhead, the conventional option produces equivalent destructive effects, and all strikes are directed at the same military target described in the scenario. Additionally, I caveat that the vignette is set in the future where all response options are developed.

	LAUNCHED FROM U.S. SOIL	LAUNCHED FROM U.S. ALLIED SOIL	LAUNCHED FROM SEA
Nuclear — Ballistic Hypersonic Glide Vehicle	Option A	Option B	Option C
Nuclear — Cruise Missile	Option D	Option E	Option F
Precision Conventional Missile	Option G	Option H	Option I

I specifically target mid-career nuclear and defense professionals, roughly ages 25–40, aiming for at least 100 participants across a variety of sectors, including the US government, government contracting, academia, think tanks, and legislative affairs/staffing. Existing experiments survey the general public or defense officials; mid-career professionals are a largely untapped network for survey experiments.⁵⁸ These professionals are the field’s future policymakers, making their perceptions of theater nuclear weapons systems directly relevant to contemporary escalation debates. Additionally, mid-career professionals have higher issue salience than the general public.

Participants anonymously choose the response option they believe is least escalatory, rank all nine options from most to least escalatory, and explain their reasoning. Upon gathering data, I will average the rankings to create an aggregate ranking of all nine options from most to least escalatory. I will also thematically code the open-ended justifications, disaggregate results by sector, and analyze the rankings both in aggregate and across groups. Beyond identifying

⁵⁸ Press, Sagan, and Valentino, “Atomic Aversion;” Lin-Greenberg, “Evaluating Escalation.”

which attributes stand out as most or least escalatory, the ranking exercise serves a second purpose: it tests whether, once given nuclear options, participants rank a nuclear system as less escalatory than a conventional one, or whether conventional options invariably occupy the least-escalatory positions. Where the nuclear options fall among the conventional provides a measure of the strength of a categorical red line around nuclear use—evidence of whether respondents reason in graduated or categorical terms. Finally, I will conduct follow-up interviews with a subset of willing participants to further explore their reasoning and thought processes in greater depth.

Wargames have become an established tool for studying decision-making in situations that one cannot observe directly.⁵⁹ By immersing participants in a hypothetical crisis and asking them to choose among response options, they generate data on how policymakers actually reason through different scenarios. Vignette-based survey experiments condense this logic into a controlled written scenario and have become a standard approach in the study of nuclear use and public and elite attitudes toward escalation.⁶⁰ Vignettes may lack the immersive depth of multi-round wargames, but they present a uniform scenario across all respondents. This standardization allows the isolation of the effect of specific weapons characteristics on perceived escalation without confounding variables of group interaction.

Contemporary scholarship on escalation perception utilizes scenario-based survey experiments to measure public opinion. Allison et al.'s *Too Brutal for War: Comparing Rationales for Weapon Taboos* and Marina E. Henke, Dino Kolonic, and Felix Lemmer's *Russian Roulette Revisited: Does a Limited Nuclear Strike Constitute a Winning Strategy?* both utilize crisis vignettes to capture participants' attention and simulate decision-making under

⁵⁹ Reid B. C. Pauly, "Would U.S. Leaders Push the Button? Wargames and the Sources of Nuclear Restraint," *International Security* 43, no. 2 (Fall 2018): 151–192, https://doi.org/10.1162/isec_a_00333.

⁶⁰ Lin-Greenberg, "Evaluating Escalation."

pressure.⁶¹ These 2026 studies legitimize and anchor the methodological choices of this study; vignettes are a common method to study such intangible topics.

These methods are not without flaws. These experiments present a perfect or pristine decision environment; no scenario can mimic the environment in which decision-makers would operate during a real crisis. Responses may also be skewed by the logic of appropriateness, with participants reporting what they believe they ought to do rather than what they would actually do.⁶² Experimental wargaming work is also limited by external validity. Results often reflect who is playing the game. While some games are run with decision makers and strategic elites, others are run with students or other professionals; conclusions drawn from one set of games may not generalize to the actual decision-makers the research is meant to inform.

Because my research question concerns perception rather than prediction, these limitations are manageable. I do not attempt to forecast crisis behavior; I seek to identify which attributes of theater nuclear weapons systems respondents associate with escalation, making a deliberately simplified vignette the most effective method. The results of this survey experiment are meant to establish which characteristics of weapons systems the broader nuclear professional community perceives as most escalatory. The trends identified here may inform current decision makers' choices about which systems the US should deploy as part of its theater nuclear force structure.

6. Next Steps

This research interrogates the assumption that escalation can be controlled with usable nuclear weapons. This summer, I have reviewed the literature and identified a gap on which to

⁶¹ Allison, Herzog, and Sukin, "Too Brutal for War" Marina E. Henke, Dino Kolonić, and Felix Lemmer, "Russian Roulette Revisited: Does a Limited Nuclear Strike Constitute a Winning Strategy?" *Contemporary Security Policy* 47, no. 2 (2026): 1–25, <https://doi.org/10.1080/13523260.2026.2624653>.

⁶² Pauly, "Would U.S. Leaders Push the Button? Wargames and the Sources of Nuclear Restraint."

focus my research: neither literature on escalation or nuclear non-use addresses the role of nuclear weapons attributes in influencing escalation. The goal of this study is to determine the significance of launch location and/or delivery vehicle on perceptions of escalation, and if the conventional-nuclear threshold is more or less significant than specific attributes. I present the following competing hypotheses:

H1: Attributes of weapons systems, independent from the carried warhead, matter in determining escalatory potential.

H1.1: Of the non-warhead attributes, the delivery vehicle has the larger effect in determining escalatory potential by compressing warning time and complicating discrimination.

H1.2: Of the non-warhead attributes, the launch location has the larger effect in determining escalatory potential by signaling that the launch site is “in play” as a legitimate target for retaliation.

H2: A weapon’s warhead type is the most significant attribute in determining escalatory potential; any conventional option is perceived as less escalatory than any nuclear option regardless of launch location or delivery vehicle.

I have written my scenario, designed my survey, and am soliciting initial feedback. After gathering data, I plan to write the following sections: results and discussion, analysis, and conclusion. In results and discussion, I will present my findings, including insights from the open-ended survey questions and places of disagreement. In the analysis section, I will apply my findings to real world scenarios, specifically related to US-Russia competition. I plan to compare and contrast my data with current conventional wisdom about theater nuclear force structure and

make my argument for which theater nuclear systems are worth investing in and why. The attributes my participants perceive to be least escalatory are incredibly valuable for determining which weapons are worth prioritizing in US theater nuclear force strategy. In my conclusion, I will synthesize my argument and offer pathways for future survey research on theater nuclear forces characteristics. For example, I am interested in examining if range (eg: short range, intermediate range, intercontinental range) and platform (eg: dual capable aircraft, mobile launchers, silos) impact perceptions of escalation.

7. Conclusion

As the US government continues to advocate for developing and deploying theater nuclear weapons, strategists and researchers must continue to question underlying assumptions behind potential nuclear use. During the Cold War era, US military strategy operated across three domains: air, land, and sea, as exemplified by the nuclear triad. Decision makers did not strongly consider strategic nuclear weapons to be realistic battlefield options and instead othered them, crediting these weapons for upholding the global deterrence regime. Today, nuclear strategy operates across five domains: air, land, sea, cyber, and outer space, and now, emerging technologies continue to blur boundaries between domains. Decision makers still consider strategic nuclear use to be unrealistic, but the promise of theater nuclear weapons to fulfill modern missions has transformed the strategic calculus and given the US new, flexible battlefield options. Theater nuclear forces are changing the relationship between nuclear weapons and strategic stability in such a way that possessing these tailored nuclear weapons seems to empower governments with a perceived ability to control escalation.

The development of these weapons is actively changing US risk perceptions and cost-benefit analyses, and decision makers require new models and frameworks to effectively strategize in an increasingly complex conflict landscape. Kahn's escalation ladder, developed in 1965, only captures a fraction of the modern warfighting experience; theater nuclear forces have transformed the battlespace into an unfamiliar domain, making different mechanisms and explanations of escalation increasingly pertinent. The escalation carousel is an attempt to add new dimensions and perspective to discussions on theater conflicts and theater nuclear use, and provide a visual model to better inform policymakers on how theater nuclear forces fit into this calculus. In contrast to Cold War frameworks, operating with a systemic rather than a state lens brings into play varied combinations of expanded warfighting domains.

Despite their existence for decades, theater nuclear weapons remain conceptually novel and underexplored relative to their influence on modern escalation and warfighting dynamics. Not all of the response options I pose in my survey yet exist, and realities of US defense budgeting and acquisition processes limit the speed at which states develop and deploy new systems. However, the underlying inquiries remain relevant: limited nuclear use would change all states' strategic calculus, so much so that escalation research on perceptions becomes not only a lens through which to study an interesting nuclear phenomenon, but also an act of nuclear risk reduction in and of itself. In modern simulations and wargames, the design frequently stops once participants consider or use nuclear weapons, rarely taking into account a post first-strike world. Should conflict ever escalate to theater nuclear use, nuclear states must be prepared to de-escalate and reduce risk— and understand how escalation dynamics change post theater use. As a critical examination of decision makers' perceptions of these weapons, this research represents the beginning of a broader strategically pertinent discussion of the frameworks nuclear

states must use to navigate escalation if the traditional conceptualization of deterrence fails and nuclear use redefines the rules of engagement.